

# Applied Deep Learning

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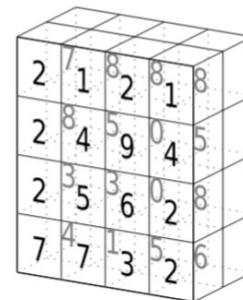
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# Linear Algebra



# Scalars, Vectors, Matrices, and Tensors

- **Scalars** are single numbers, e.g.,  $x \in \mathbb{R}, n \in \mathbb{N}$
- **Vectors** are arrays of numbers,  $x = [0.2, 1, 5.4]^T \in \mathbb{R}^3$
- **Matrices** are 2D arrays of numbers,  $A = \begin{bmatrix} 4 & -.1 \\ .3 & .6 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$
- **Tensors** are kD arrays of numbers,  $A =$



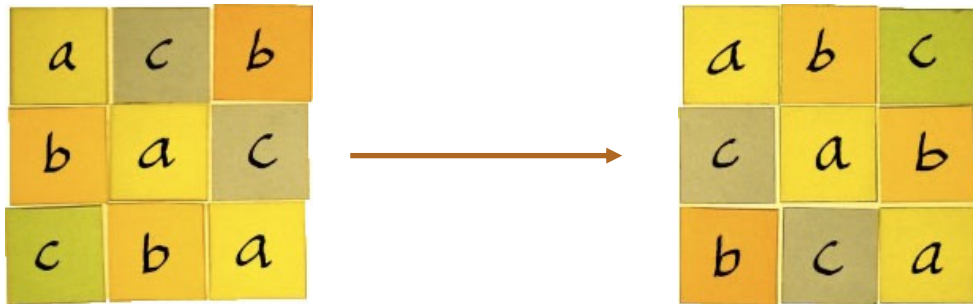
|   |   |   |   |
|---|---|---|---|
| 2 | 1 | 8 | 8 |
| 2 | 8 | 5 | 0 |
| 2 | 5 | 3 | 6 |
| 7 | 7 | 1 | 3 |



# Algebraic Operations

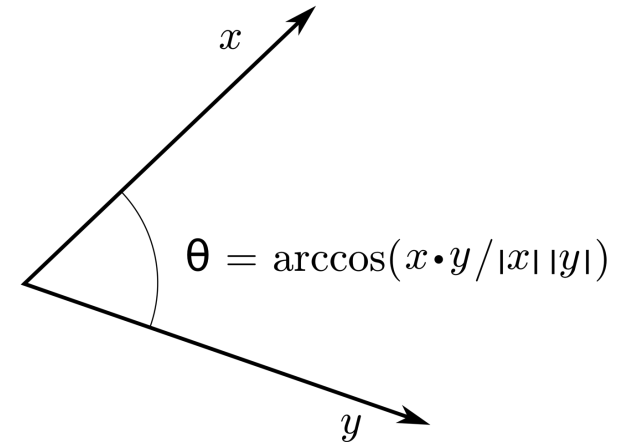
## Transpose

$$A_{ij}^T = A_{ji}$$



## Dot product

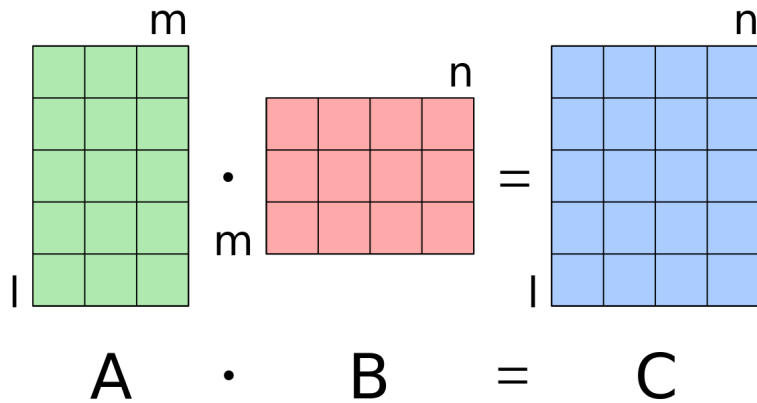
$$\langle x, y \rangle = x^T y = \sum_j x_j y_j$$



# Algebraic Operations

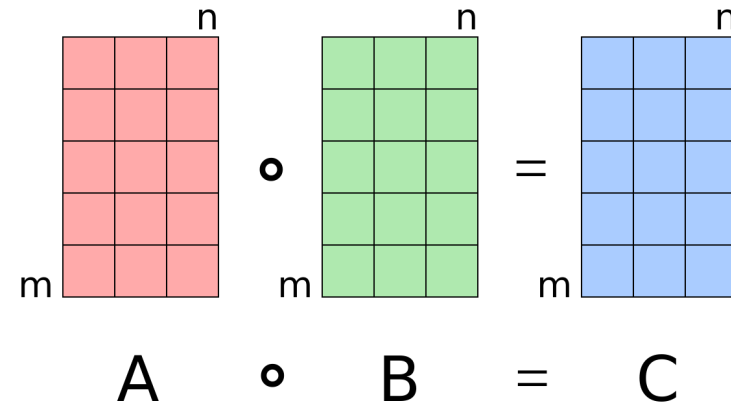
## Matrix Product

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}$$



## Hadamard Product

$$(A \circ B)_{ij} = A_{ij} B_{ij}$$



# System of Linear Equations

Let  $A \in \mathbb{R}^{m \times n}$  and  $b \in \mathbb{R}^m$

$$Ax = b \quad \longrightarrow \quad \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$



# Identity and Inverse Matrices

## Identity

$$Ix = x$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

## Inverse

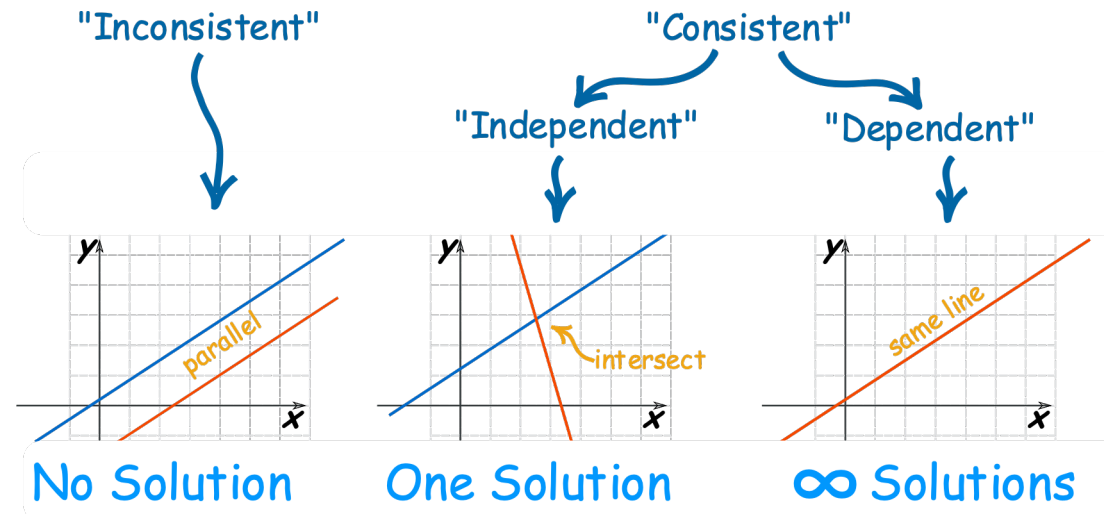
$$Ax = b \implies x = A^{-1}b$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$$



# Solutions of Linear Systems

1. The system  $Ax = b$  has a **solution** for all  $b \in \mathbb{R}^m$  if  $A \in \mathbb{R}^{m \times n}$  spans an  $m$ D space
2. The matrix inverse exists if  $A$  is **square**, i.e.,  $m = n$





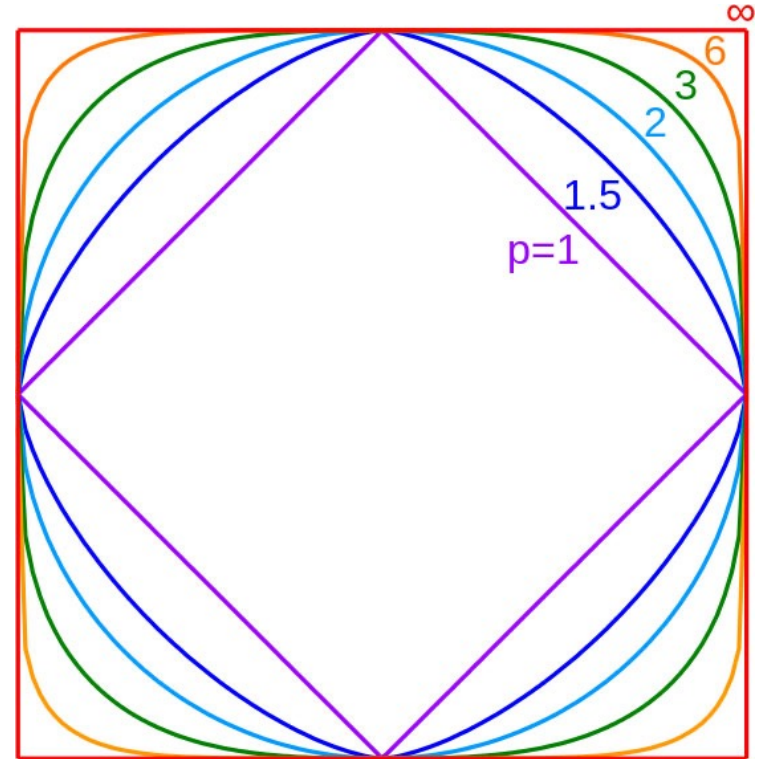
# Vector Norms

The  $L^p$  norm

$$|x|_p = \left( \sum_j |x_j|^p \right)^{\frac{1}{p}}$$

The  $L^\infty$  norm

$$|x|_\infty = \max_j |x_j|$$



# Matrix Norms

The Frobenius norm

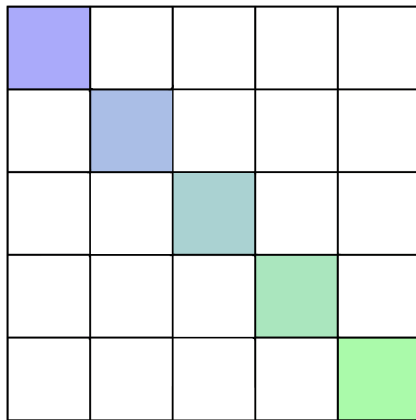
$$|A|_F = \left( \sum_{ij} A_{ij}^2 \right)^{\frac{1}{2}}$$

- We have  $|A|_F = |\text{vec}(A)|_2 = \text{Tr}(A^T A)^{\frac{1}{2}}$

# Special Matrices

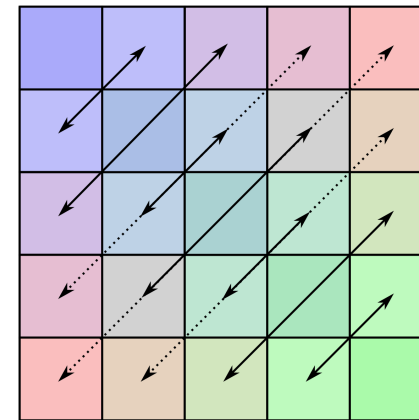
## Diagonal

$$D_{ij} = \begin{cases} d_i & i = j \\ 0 & i \neq j \end{cases}$$



## Symmetric

$$A = A^T$$



# Eigendecomposition

Given a square  $A$  with  $v \neq 0$

*eigenvector*



$$Av = \lambda v$$



*eigenvalue*

Typically,  $|v| = 1$

Written differently

$$AV = V\Lambda$$

If  $A$  is *symmetric*

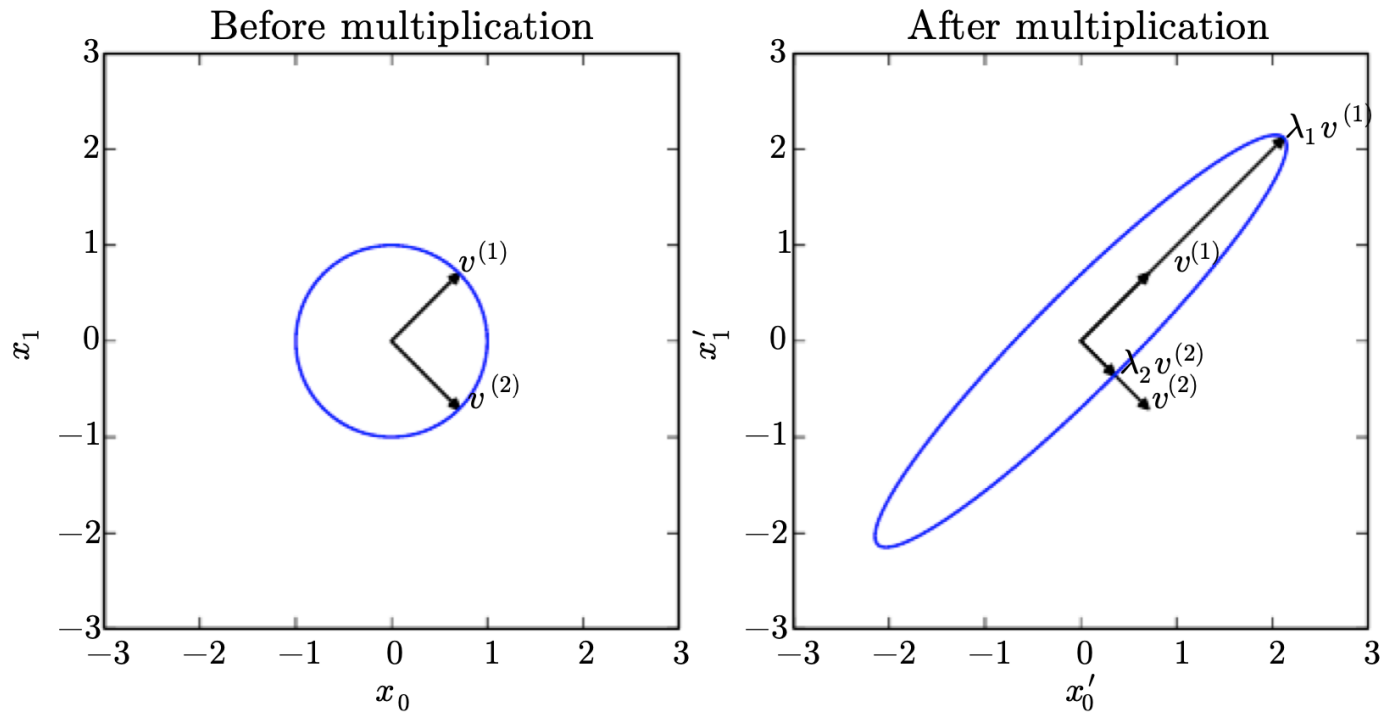
$$A = Q\Lambda Q^T$$



*orthogonal*



# Eigendecomposition



# Singular Value Decomposition

For every  $A \in \mathbb{R}^{m \times n}$

singular values

(left) singular vectors

(right) singular vectors

$$A = \underset{m \times m}{U} \underset{m \times n}{S} \underset{n \times n}{V}^T$$

- Eigendecomposition and SVD are **related**



# Moore—Penrose Pseudoinverse

No inverse for **non-square**  $A$ !

MP **pseudoinverse**

$$A^+ = \lim_{\alpha \rightarrow 0} (A^T A + \alpha I)^{-1} A^T$$

In practice

$$A^+ = V S^+ U^T$$

where  $S^+ = (S^{-1})^T$

↑  
reciprocal of  
nonzeros

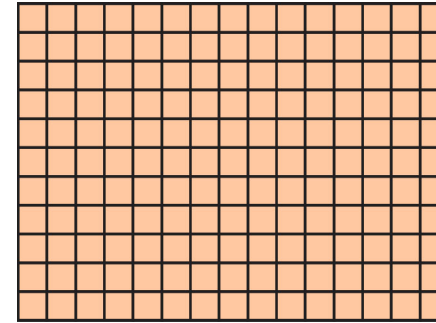


# Moore—Penrose Pseudoinverse

$$Ax = b \implies x = A^+ b$$



$$|Ax - b|_2$$



$$|x|_2$$





# Example: Principal Component Analysis

Given a collection of  $n$  points  $\{x^{(1)}, \dots, x^{(n)}\} \in \mathbb{R}^m, m \leq n$

$$x \xrightarrow{f} \underset{k}{c} \xrightarrow{g} y \approx x$$

How can this be achieved?



# Example: Principal Component Analysis

Denote

$$X = (x^{(1)} \quad \dots \quad x^{(n)})$$

Recall the SVD

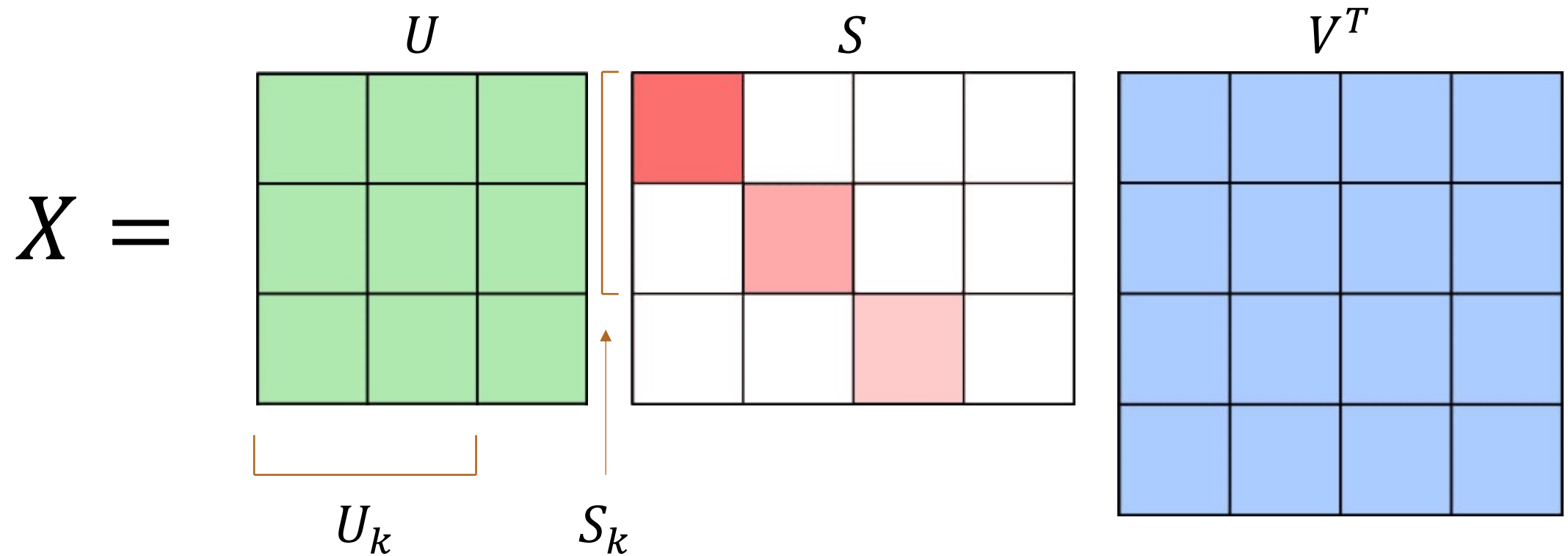
$$X = U S V^T$$

orthogonal

nonnegative



# Example: Principal Component Analysis



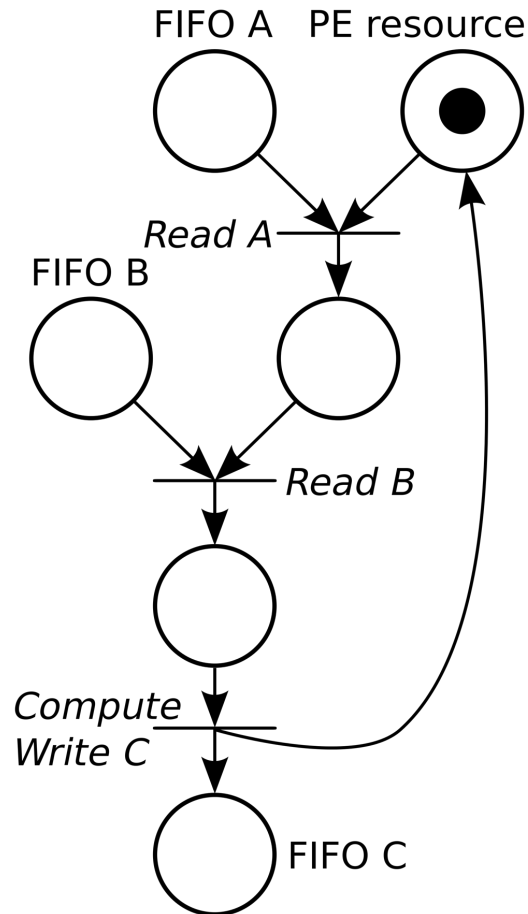
# Code Demo



# Probability



# Motivation – Probability

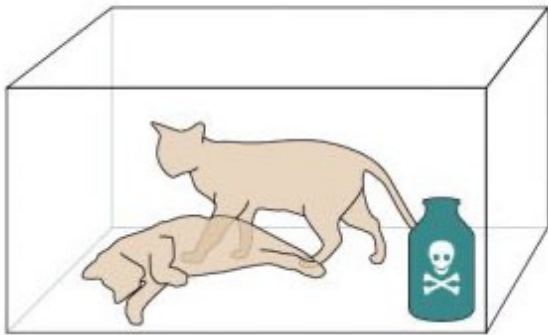


Why computer scientists would need **probability theory**?

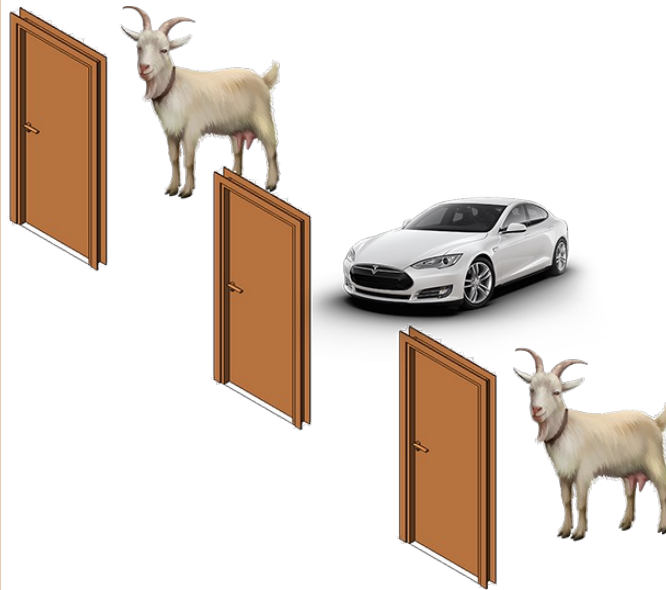


# Sources of Uncertainty

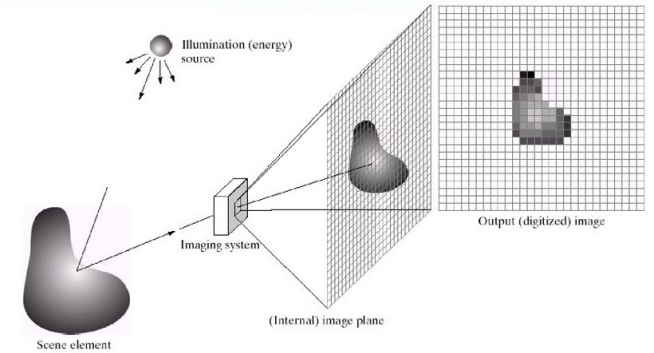
## Inherent Stochasticity



## Incomplete Observability



## Incomplete Modeling



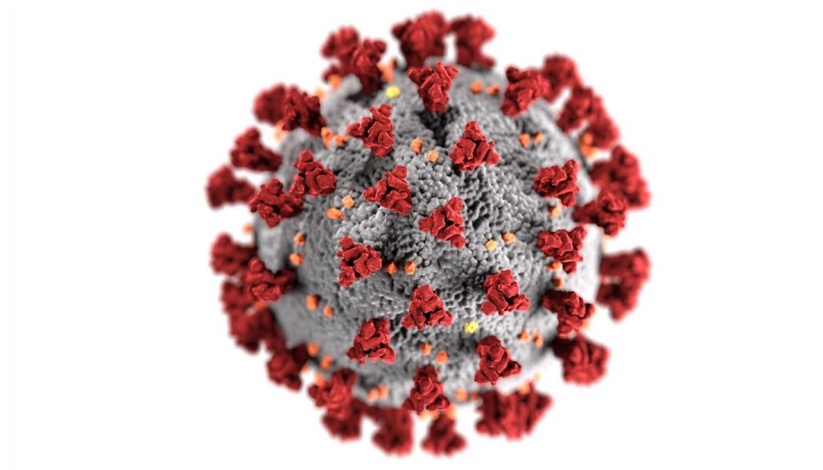
# Artificial Intelligence and Probability

Frequentist probability



1:220

Bayesian probability



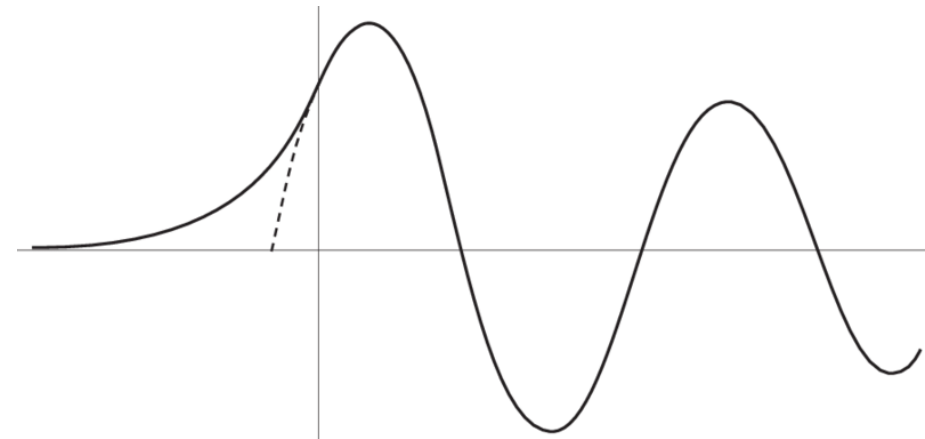


# Random Variables

## Discrete



## Continuous



# Probability Distribution

The **likelihood** of a random variable to take on each of its possible states

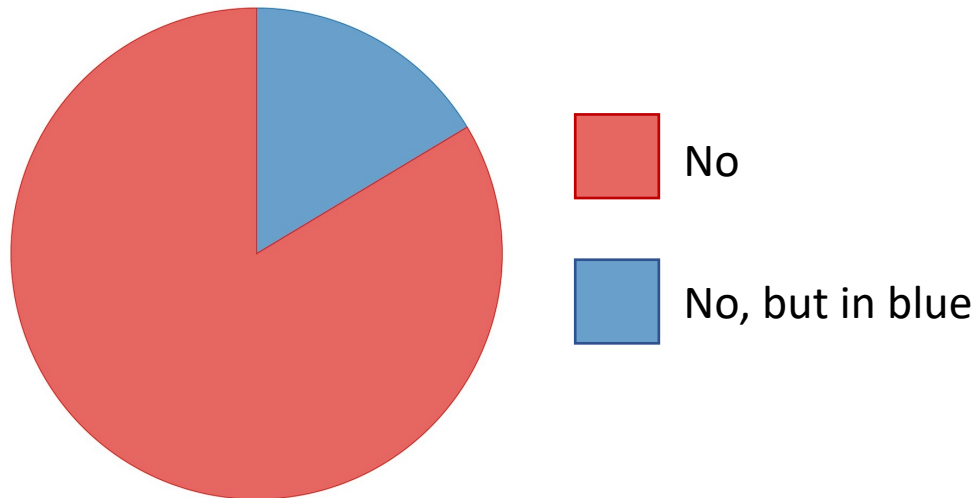


# Probability Mass Function (PMF)

Single Variable

$$P(x = x)$$

Are those pie charts memes  
ever funny?



Multiple Variables

$$P(x = x, y = y)$$

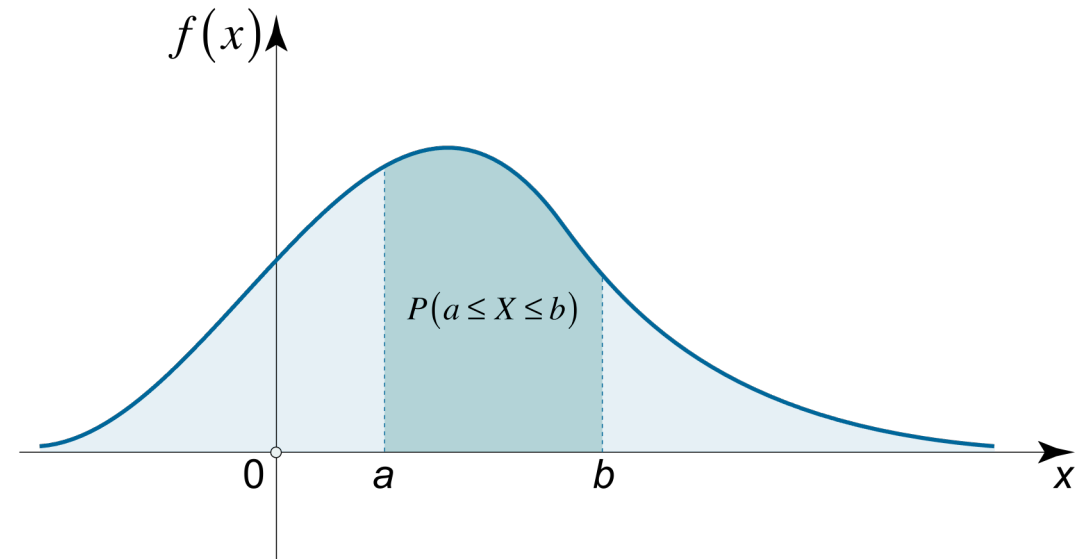
↑  
joint probability  
distribution



# Probability Density Function (PDF)

No probability for **specific** states!

$$\int_{[a,b]} p(x)dx$$



# Marginal Probability Distribution

PD over a **subset** of variables

Discrete:

$$P(x = x) = \sum_y P(x = x, y = y)$$

Continuous:

$$p(x) = \int p(x, y) dy$$

*sum rule*

|         | $y = 5$ | $y = 6$ |
|---------|---------|---------|
| $x = 1$ | 0.2     | 0.3     |
| $x = 2$ | 0.4     | 0.1     |



# Conditional Probability

probability of an event, given that some **other** event has happened

$$P(y = y \mid x = x) = \frac{P(y = y, x = x)}{P(x = x)}$$

|         | $y = 5$ | $y = 6$ |
|---------|---------|---------|
| $x = 1$ | 0.2     | 0.3     |
| $x = 2$ | 0.4     | 0.1     |



# Chain Rule of Conditional Probabilities

Any joint PD may be **decomposed** into

$$P(x^{(1)}, \dots, x^{(n)}) = P(x^{(1)}) \prod_{i=2}^n P(x^{(i)} \mid x^{(1)}, \dots, x^{(i-1)})$$



# Independence and Conditional Independence

$x$  and  $y$  are **independent** if

$$p(x, y) = p(x)p(y)$$

for every  $x \in x$  and  $y \in y$

$x$  and  $y$  are **conditionally independent** given  $z$  if

$$p(x, y | z) = p(x | z)p(y | z)$$

for every  $x \in x$ ,  $y \in y$  and  $z \in z$





# Example: Independent Random Variables

The uniform distribution

|         | $y = 5$ | $y = 6$ |
|---------|---------|---------|
| $x = 1$ | 0.25    | 0.25    |
| $x = 2$ | 0.25    | 0.25    |



# Expectation

The **expected value** of  $f(x)$

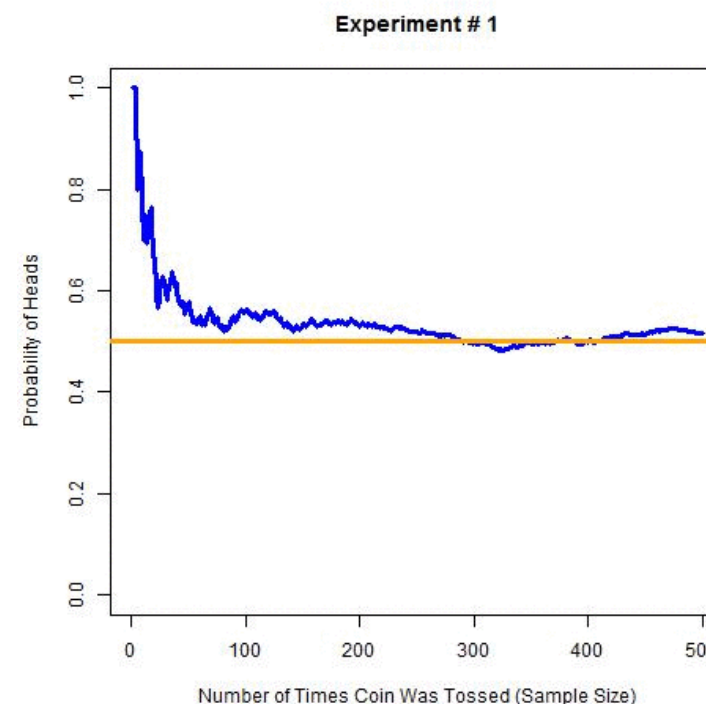
Discrete:

$$\mathbb{E}_{X \sim P}[f(x)] = \sum_x P(x)f(x)$$

Continuous:

$$\mathbb{E}_{X \sim P}[f(x)] = \int p(x)f(x)dx$$

↑  
*linear*



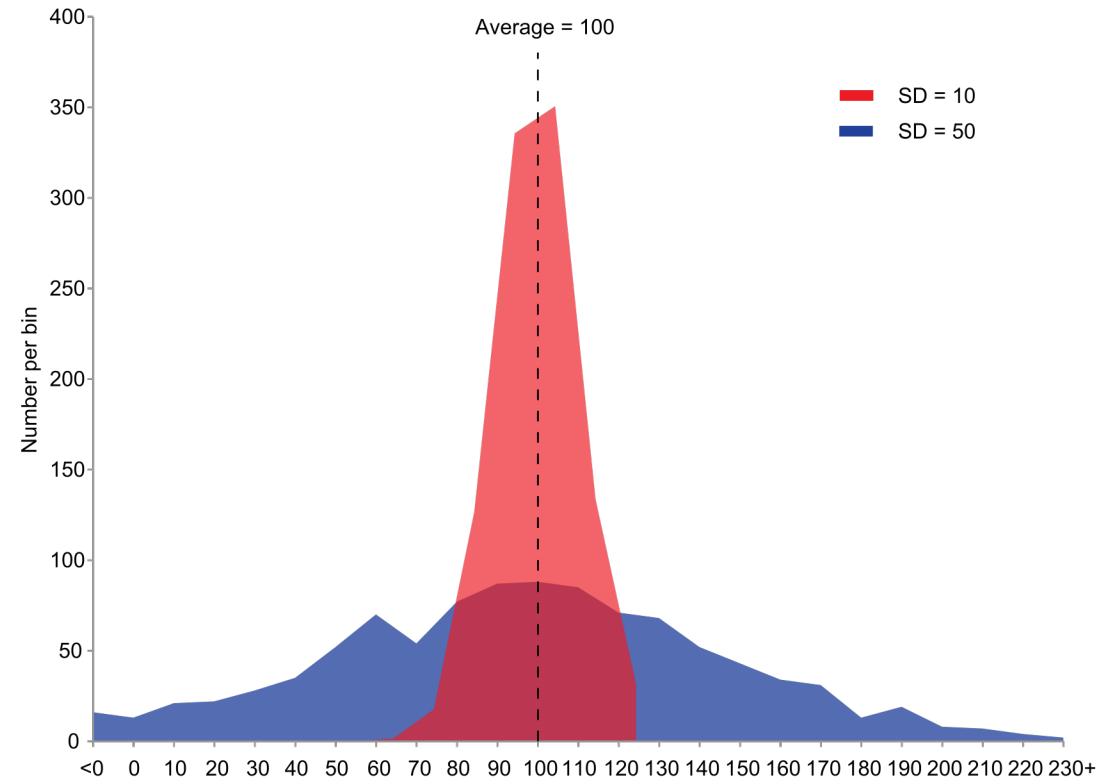
# Variance

the **difference** between  $f(x)$  and its expectation

$$\text{Var}(f(x)) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2]$$

$(\cdot)^{\frac{1}{2}}$

standard  
deviation



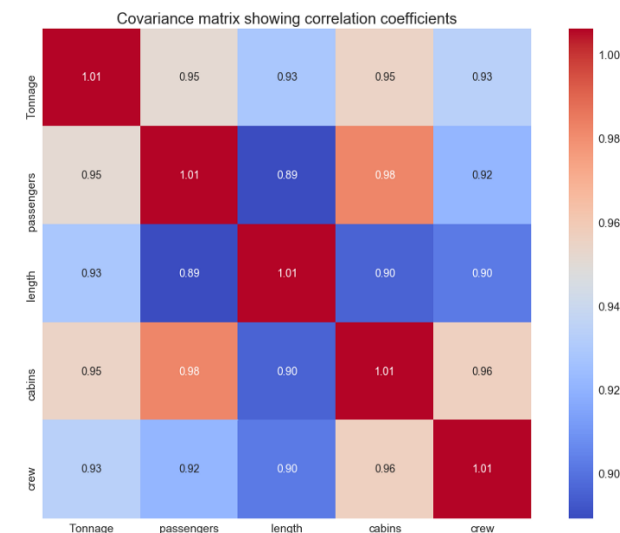
# Covariance

Measures the **linear** relationship between  $f(x)$  and  $g(y)$

$$\text{Cov}(f(x), g(y)) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])(g(y) - \mathbb{E}[g(y)])]$$

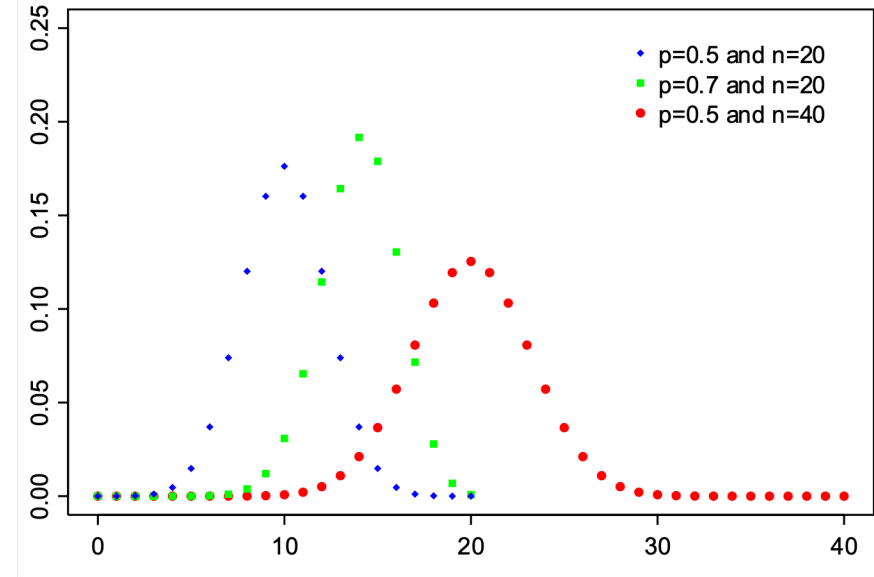
The covariance matrix of  $x$  is

$$\text{Cov}(x)_{ij} = \text{Cov}(x_i, x_j)$$



# Common Probability Distributions

## Binomial distribution



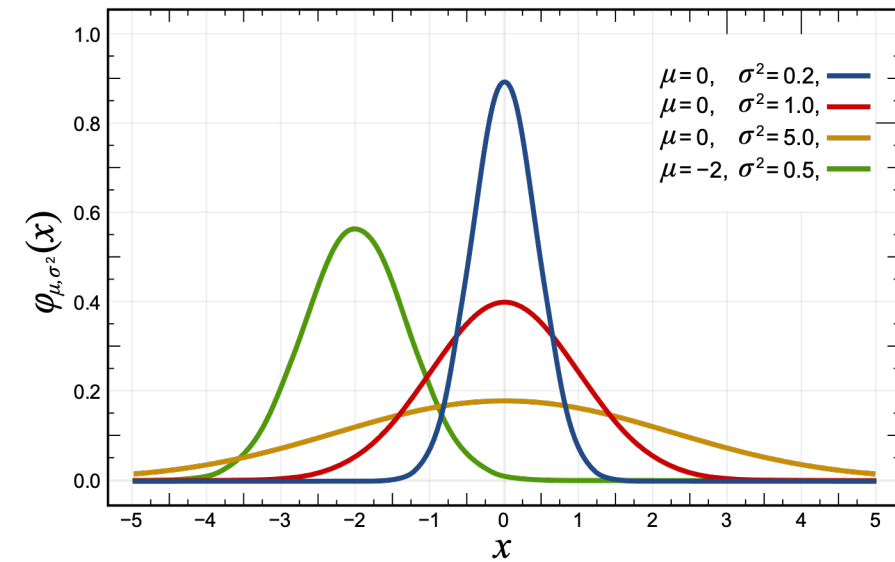
$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$



# Common Probability Distributions



## Gaussian distribution



$$\mathcal{N}(x; \mu, \sigma^2) = \sqrt{\frac{1}{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

↑  
mean & variance



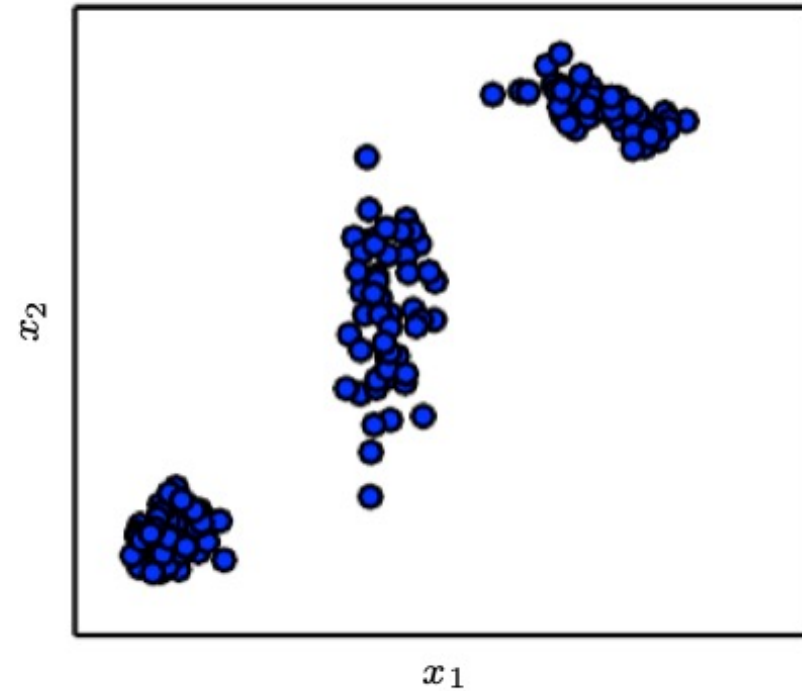
# Example: Gaussian Mixture Models

Consider the mixture distribution

$$P(\mathbf{x}) = \sum_j P(c = j) P(\mathbf{x} | c = j)$$

prior probability

gaussians  
 $\mu^{(j)}, \sigma^{(j)}$



# Common Functions

logistic sigmoid

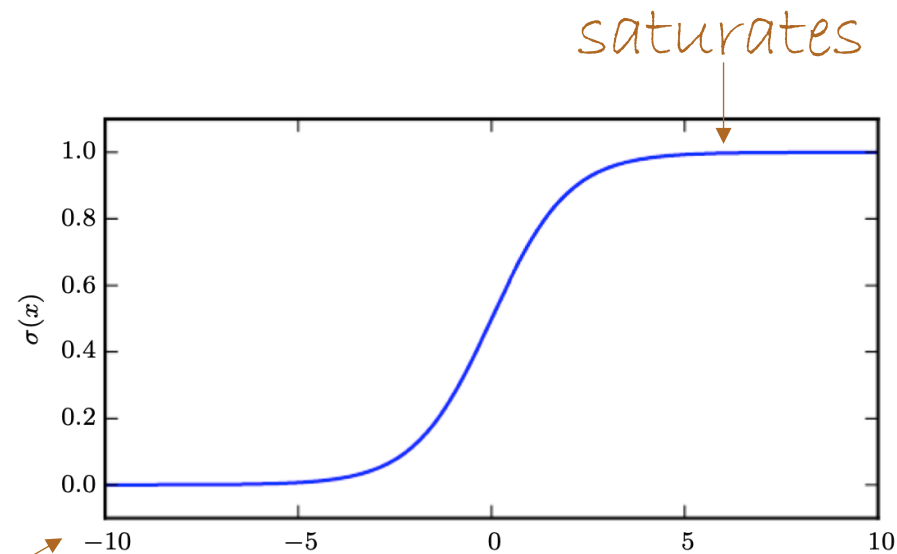


Figure 3.3: The logistic sigmoid function.

$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$

softplus

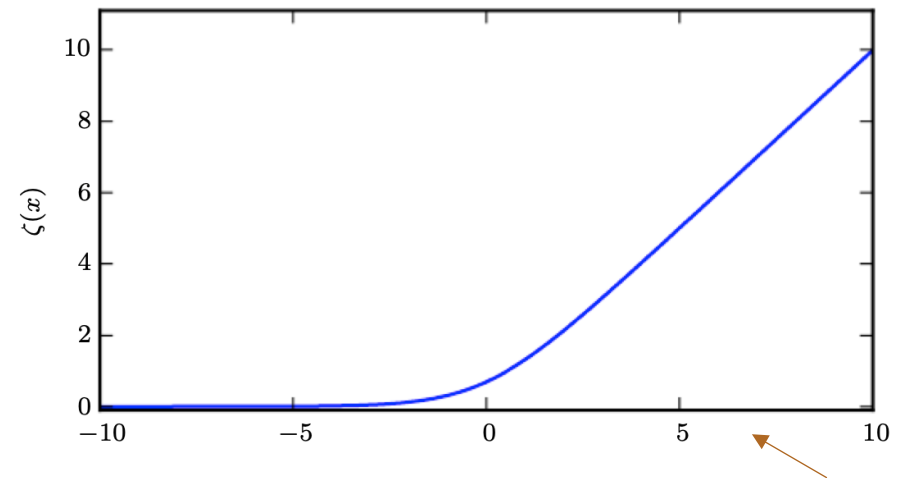


Figure 3.4: The softplus function.

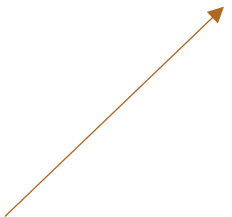
$$\zeta(x) = \log(1 + \exp(x))$$

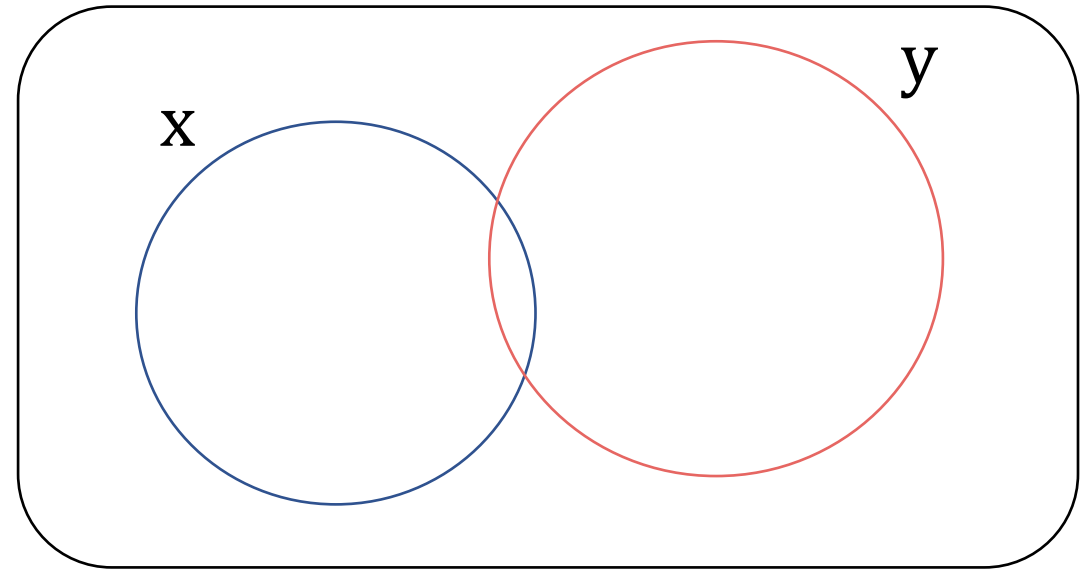




# Bayes' Rule

$$P(x | y) = \frac{P(x)P(y | x)}{P(y)}$$

$$P(y) = \sum_x P(y | x)P(x)$$




# Distance between Distributions

Kullback—Leibler divergence

$$D_{KL}(P\|Q) = \mathbb{E}_{x \sim P} \left[ \log \frac{P(x)}{Q(x)} \right]$$

↑  
*asymmetric*

